Reliability Study of RD-Agent-Proposed Approaches to 12-Hour Acute Respiratory Failure Prediction

Nested Cross-Validation, Statistical Uncertainty,
Probability Calibration, and Threshold-Policy Analysis

English translation of the corresponding Chinese research report.

Abstract

Research Background and Objective

This project audits and validates acute respiratory failure (ARF) prediction approaches proposed by RD-Agent, an automated research-and-development agent. Using ICU clinical time-series data, it predicts whether ARF will occur in the 12 hours following a 12-hour observation window.

This Final Reliability Report is the project's current formal reliability evaluation. It supersedes ARF V1 for formal performance assessment; ARF V1 remains an exploratory record, and its metrics arise from a different validation protocol.

The aim is not to establish clinical deployability, but to determine whether the modeling directions developed by RD-Agent remain valid under a stricter, independent, and reproducible experimental framework. The formal analysis includes a fixed outer split, nested parameter selection, fair model comparison, statistical uncertainty analysis, cross-fitted probability calibration, and threshold-policy analysis.

The report reuses the publicly released `arf-12-hours-prediction-task` sample resource used in ARF V1. The available task resource contains preprocessed, anonymized ICU-stay-level features and does not provide patient-level identifiers. This report does not independently document the upstream clinical database or release version, cohort-selection rules, or the operational process used to construct the ARF label; it therefore evaluates methodological reliability on the supplied task resource rather than clinical validity for a separately traceable patient cohort.

Data and Experimental Design

The dataset contains 9,921 samples:

- Positive samples: 960
- Negative samples: 8,961
- Positive rate: 9.68%
- Input observation window: 12 hours
- Raw features per hour: 4,816

The formal experiment uses a fixed five-fold stratified outer split. Within each outer-training set, three-fold stratified cross-validation selects model parameters. Logistic regression and LightGBM use exactly the same outer folds and the same 12-hour sum features. Outer-validation labels do not participate in inner-loop parameter selection.

AUROC is the primary metric, with Average Precision (AP) and Brier score as secondary metrics. Key inputs, data splits, results, and reports are documented and verified using SHA-256 hashes, and automated tests recheck metrics and output structure.

Primary Predictive Results

Model	Mean fold AUROC	Pooled OOF AUROC	Pooled OOF AP	Pooled OOF Brier score
Logistic regression	0.7080	0.7077	0.2288	0.1515
LightGBM	0.7758	0.7537	0.3017	0.0887

LightGBM outperformed logistic regression on AUROC and AP in all five outer folds. The mean paired AUROC difference across the five folds was +0.0677, with a 95% confidence interval of [0.0298, 0.1056] obtained using an approximate correction.

Because there were only five outer-fold pairs, the two-sided exact sign test gave p = 0.0625. A supplementary stratified bootstrap also supported LightGBM. However, it resampled fixed OOF predictions without repeating full model training and parameter selection, so it is only supplementary descriptive evidence.

Mean fold AUROC and pooled OOF AUROC are not the same statistic. Pooled OOF predictions are produced by five separately trained models, whose probability scales may differ across folds. Both results are retained and are not conflated.

Probability Calibration and Threshold Results

Class-weighted models systematically overestimated the probability of ARF. Cross-fitted Platt scaling improved the reliability of predicted probabilities without fitting the calibrator or selecting the threshold using labels from the target outer fold.

For LightGBM, the pooled OOF results after Platt scaling were:

- Brier score: decreased from 0.0887 to 0.0790
- Log loss: decreased from 0.3030 to 0.2808
- Expected calibration error computed over 10 quantile bins: decreased from 0.0654 to 0.0116
- Mean predicted probability: decreased from 15.85% to approximately 9.88%, close to the 9.68% event rate

Because each target fold uses a different calibrator, the overall ranking across folds can change after pooling even when the Platt mapping is monotonic within each fold. Therefore, pooled OOF AUROC after cross-fitted calibration need not remain exactly unchanged.

Threshold analysis showed:

- A fixed threshold of 0.5 gave approximately 25.9% sensitivity and approximately 95.8% specificity
- The 0.80 target-sensitivity policy achieved approximately 78.4% realized sensitivity on unseen outer folds while classifying approximately 49.4% of samples as positive
- The 0.90 target-sensitivity policy achieved approximately 89.2% realized sensitivity while classifying approximately 65.7% of samples as positive

There is therefore no single optimal threshold independent of the costs of missed diagnoses and false positives, review capacity, and other resource constraints.

Main Conclusions

Under the current data, fixed 12-hour sum features, and strict nested cross-validation framework, LightGBM shows a consistent and practically meaningful predictive advantage over logistic regression.

Cross-fitted Platt scaling improves the reliability of predicted probabilities, but internal calibration cannot replace external hospital calibration. Increasing sensitivity substantially raises the predicted-positive proportion and false-positive burden, so threshold policy must be set in light of actual costs and resource constraints.

The conclusions are limited to the current retrospective data and experimental framework. They do not demonstrate clinical validity, generalizability, or deployment feasibility.

Main Limitations

- No independent external-hospital validation
- No patient-level identifiers to fully audit leakage across admissions for the same patient
- Only five outer folds, limiting fold-level statistical comparisons
- Anonymized feature names, limiting clinical interpretation
- Internal cross-fitted calibration only
- No real medical-cost, bed-capacity, or manual-review-capacity data for threshold analysis
- No prospective clinical workflow or patient-outcome evaluation

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Part I. Rigorous Nested Cross-Validation and Statistical Uncertainty Analysis

1. Research Purpose

This experiment uses nested cross-validation to compare logistic regression and LightGBM under a fixed, reproducible data split. It prevents the outer validation folds from being used for hyperparameter selection and evaluates the stability and uncertainty of the observed difference in performance between the two models.

This experiment focuses on answering the following questions:

- Does LightGBM outperform logistic regression under the same data, features, and outer-fold split?
- Is this performance advantage consistent across the five outer validation folds?
- Are parameter selection and final evaluation strictly segregated?
- How much uncertainty is there in the observed performance differences?
- Are the results sufficient to support the model's superiority on the current data, and what are the limitations?

2. Relationship Between ARF V2 and RD-Agent

ARF V1 used RD-Agent to automatically generate and run an ARF prediction pipeline, demonstrating that the automated data-science agent could complete feature construction, model training, cross-validation, and submission-file generation.

ARF V2 does not rerun RD-Agent directly. Instead, it provides an independent, rigorous, and auditable review of the modeling approach developed in ARF V1.

Independent experimental code in ARF V2 performs the following work:

- Fixing the outer-fold data split
- Building a unified data loader
- Developing automated tests
- Performing nested cross-validation
- Auditing inner-loop parameter selection
- Recalculating outer-fold and OOF metrics
- Conducting paired statistical and uncertainty analyses
- Saving configurations, logs, results, and SHA-256 hash records

Accordingly, this stage is not intended to attribute the results of manually conducted experiments to RD-Agent; rather, it evaluates and validates whether the modeling directions generated by the automated agent withstand more rigorous experimental scrutiny.

3. Data

3.1 Task Definition

The task is to predict whether the sample will develop acute respiratory failure (ARF) in the next 12 hours based on the clinical time series of the past 12 hours of each ICU stay.

This is a class-imbalanced binary classification task.

3.2 Data Scale

- Number of samples: 9,921
- Positive samples: 960
- Negative samples: 8,961
- Positive rate: 9.68%
- Input time window: 12 hours
- Raw features per hour: 4,816
- Original tensor shape: `(9921, 12, 4816)`

3.3 Feature Representation

This experiment sums each raw feature over 12 hours, converting the three-dimensional time series tensor into a two-dimensional sparse feature matrix.

The transformed feature matrix shape is:

```
(9921, 4816)
```

This feature representation is recorded as:

```
12h_sum
```

The reason for choosing the summation feature is that ablation experiments and multiple random seed experiments on ARF V1 show that the 12-hour summation feature has similar or better predictive performance than the 43,344-dimensional full feature containing nine categories of temporal statistics, while being substantially less computationally expensive.

It should be noted that this feature selection refers to the results of previous experiments. Therefore, the goal of ARF V2 is to rigorously compare the two models rather than to provide a completely unbiased final evaluation of all possible feature engineering scenarios.

3.4 Data Integrity

The original training data includes:

- `X.npz`
- `ARF_12h.csv`

SHA-256 hashes were recorded for both files and verified again before the experiment began. The verification results were:

```
OK
```

Therefore, the data used in this experiment are consistent with the data used when the experimental archive was established.

Raw medical data are not included in the public research repository. The repository contains only SHA-256 hash records, data-split files, experimental code, summary results, and analysis reports.

4. Experimental Environment

The experimental operating environment is:

- Operating system: Ubuntu 24.04 LTS
- CPU architecture: ARM64/aarch64
- Virtual machine memory: approximately 5.8 GiB
- Swap: approximately 3.8 GiB
- Python: 3.10.20
- NumPy: 2.2.6
- pandas: 2.3.3
- SciPy: 1.14.1
- scikit-learn: 1.7.2
- LightGBM: 4.7.0
- RD-Agent: source code installation

Environmental information is stored in:

```
archive/arf_v1/environment.txt
```

5. Fixed Outer-Fold Partition

The fixed outer-fold split is generated locally from the released configuration. The sample-level split file is intentionally not included in the public repository; its SHA-256 record is retained for authorized reproduction.

The partitioning method is:

```
StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=42,
)
```

The data-split file contains the following fields:

- `sample_index`
- `ID`
- `label`
- `outer_fold`

Integrity check results:

- Total number of records: 9,921
- Unique `sample_index`: 9,921

- Unique ID: 9,921
- Outer-fold identifiers: 0, 1, 2, 3, 4
- Number of samples in each fold: 1,984-1,985
- Positive rate of each fold: about 9.68%

The SHA-256 hash of the fixed data-split file is:

```
265ec7b207e85a6bfcf12965d7b7a701b60d459ce64fff95c0f52a77bad95903
```

Fixed partitioning enables all models to use the exact same outer training and validation sets, thereby improving the fairness and reproducibility of model comparisons.

6. Experimental Design

6.1 Nested Cross-Validation

This experiment uses nested cross-validation:

- Outer loop: fixed five-fold stratified cross-validation for final performance evaluation
- Inner loop: three-fold stratified cross-validation, used only for hyperparameter selection
- Outer random seed: 42
- Inner random seed: 2026
- Primary evaluation metric: AUROC
- Secondary evaluation metrics: Average Precision and Brier score

The outer validation set does not participate in hyperparameter selection.

The procedure for each outer fold is:

```
Outer training set
    ↓
Select hyperparameters by three-fold stratified inner cross-validation
    ↓
Refit on the full outer training set using the best hyperparameters
    ↓
Generate predicted probabilities for the outer validation set
    ↓
Save that fold’s performance metrics and OOF predictions
```

6.2 Logistic regression

Logistic regression settings include:

- `solver="liblinear"`
- `class_weight="balanced"`
- `max_iter=2000`

The regularization parameters for the inner cross-validation search are:

```
C ∈ {0.01, 0.1, 1.0, 10.0}
```

The selected value of C was identical across all five outer folds:

```
C = 0.01
```

6.3 LightGBM

The basic settings of LightGBM include:

- Binary classification target
- AUROC as the inner-loop hyperparameter-selection metric
- Learning rate: 0.02
- Maximum number of boosting rounds: 3,000
- Early stopping: 100 rounds
- Class weights are calculated from the negative-to-positive ratio in the corresponding training data.
- Fixed random seeds control the training process
- Number of CPU threads: 4

The parameter grid includes the following four combinations:

num_leaves	min_data_in_leaf	feature_fraction	bagging_fraction
31	50	0.8	0.8
31	100	0.8	0.8
63	50	0.8	0.8
63	100	0.8	0.8

The final number of boosting rounds is determined from the optimal iteration count in the inner validation folds for the best parameter combination.

The final numbers of boosting rounds used in the five outer folds were:

Outer fold	Boosting rounds
0	368
1	440
2	381
3	351
4	282

This experiment used a pre-specified finite parameter grid and did not search the entire parameter space of LightGBM. Therefore, the results represent a comparison of the two models within the current preset search range, rather than a demonstration of the theoretically optimal performance of each model.

7. Automated Testing and Integrity Checking

Automated tests were established and run before and after model execution, covering:

- Original tensor structure

- Label count and class distribution
- ID uniqueness
- Fixed-partition integrity
- Alignment among the data, labels, and fixed data-split ordering
- Correspondence between sparse-tensor coordinates and values
- Shape of the 12-hour summary-feature matrix
- Validity of the experimental configuration
- Outer- and inner-loop cross-validation settings
- Completeness of OOF predictions
- Range of prediction probabilities
- Outer-fold integrity
- Recalculation of AUROC, AP, and Brier score for each fold
- Recalculation of pooled OOF metrics
- Whether the best hyperparameters selected in the inner loop indeed achieved the highest mean inner-loop AUROC

25 automated tests passed during the nested-cross-validation stage. After probability-calibration and threshold-policy analyses were added, the total number of automated tests in the complete project increased to 32.

```
25 passed
```

The complete nested-cross-validation run generated the following core result files:

- `nested_cv_fold_results.csv`
- `nested_cv_summary.csv`
- `nested_cv_oof_predictions.csv`
- `nested_cv_selected_parameters.csv`
- `arf_nested_cv_v1.log`

The public release includes only `nested_cv_fold_results.csv`, `nested_cv_summary.csv`, and `nested_cv_selected_parameters.csv`; their SHA-256 records pass verification. The OOF prediction file and complete execution log were generated but are excluded because they contain individual-level predictions and the complete execution record, respectively.

8. Nested Cross-Validation Results

8.1 Results by Outer Fold

Model	Outer fold	AUROC	Average Precision	Brier score
Logistic Regression	0	0.6897	0.1970	0.1541
LightGBM	0	0.7616	0.2611	0.0846
Logistic Regression	1	0.6996	0.2294	0.1539
LightGBM	1	0.7565	0.2914	0.0823
Logistic Regression	2	0.7039	0.2229	0.1503
LightGBM	2	0.7872	0.3742	0.0761
Logistic Regression	3	0.7350	0.2563	0.1420
LightGBM	3	0.7734	0.3345	0.0967
Logistic Regression	4	0.7119	0.2691	0.1571
LightGBM	4	0.8000	0.3393	0.1037

8.2 Mean Results Across Folds

Model	Average AUROC	AUROC standard deviation	Average AP	AP standard deviation
Logistic Regression	0.7080	0.0171	0.2349	0.0285
LightGBM	0.7758	0.0180	0.3201	0.0442

Across all five outer folds, LightGBM achieved higher AUROC and Average Precision than logistic regression.

8.3 Pooled OOF Results

Model	OOF AUROC	OOF AP	OOF Brier score
Logistic Regression	0.7077	0.2288	0.1515
LightGBM	0.7537	0.3017	0.0887

LightGBM's mean AUROC across the outer folds is 0.7758, whereas its pooled OOF AUROC is 0.7537.

These two values are generated by different calculation methods:

- Mean AUROC across the outer folds: compute AUROC separately for each of the five outer folds, then take the arithmetic mean.

- Pooled OOF AUROC: concatenate OOF predictions from the five outer-fold models, then compute AUROC once across all samples.

Because the five outer-fold models are trained independently, their predicted-probability scales may differ. When these probabilities are concatenated, the overall ranking of samples across folds may change. Therefore, the mean AUROC across the folds and the pooled OOF AUROC need not be the same.

This report retains both metrics and does not refer to OOF AUROC as the “five-fold mean AUROC.”

A lower Brier score indicates a smaller squared error between predicted probabilities and true labels. The basic nested cross-validation results above have not yet undergone post-processing calibration. Therefore, the Brier score here can only be used as a supplementary metric and cannot by itself demonstrate that the probabilities have already been well calibrated.

After completing the basic model comparison, this project further conducted probability calibration analysis using cross-fitted Platt scaling and isotonic regression. The relevant public code, aggregate tables, figure, and automated check are listed in Section 11.1.

9. Paired Comparison Across the Five Outer Folds

The AUROC improvement of LightGBM relative to logistic regression is:

Outer fold	AUROC difference
0	+0.0719
1	+0.0568
2	+0.0834
3	+0.0384
4	+0.0881

The improvements in Average Precision are:

Outer fold	AP difference
0	+0.0641
1	+0.0621
2	+0.1512
3	+0.0781
4	+0.0702

The mean paired difference is:

- AUROC: `+0.0677`
- Average Precision: `+0.0852`

LightGBM achieved positive improvements in both ranking metrics across the five outer folds, indicating strong directional consistency.

10. Statistical Uncertainty Analysis

10.1 Conventional Paired Confidence Intervals

Based on paired differences across the five outer folds, the conventional t-interval results are:

Metric	Mean difference	Conventional 95% interval
AUROC	0.0677	[0.0424, 0.0930]
Average Precision	0.0852	[0.0386, 0.1317]

The conventional confidence interval treats the five fold-level differences as approximately independent observations. Because the cross-validation training sets overlap, it may underestimate uncertainty.

10.2 Corrected Cross-Validation Confidence Intervals

After applying an approximate correction to account for training-set overlap, the results are:

Metric	Mean difference	Corrected 95% interval
AUROC	0.0677	[0.0298, 0.1056]
Average Precision	0.0852	[0.0154, 0.1549]

Neither corrected interval crosses zero, providing consistent evidence for the performance advantage of LightGBM.

This correction is approximate and is based on a single five-fold cross-validation; it relies on statistical assumptions and cannot substitute for independent external validation.

10.3 Exact Sign Test

LightGBM outperformed logistic regression on both AUROC and AP in all five outer folds.

The result of the two-sided exact sign test is:

```
p = 0.0625
```

Since there are only five pairs of observations, when the five differences are all in the same direction, the minimum p-value that can be achieved with a two-sided sign test is 0.0625.

Therefore, this result indicates that the model advantage has a consistent direction across the five folds, but does not meet the traditional `p < 0.05` criterion for significance. It cannot be stated as "significantly better by sign test".

10.4 Stratified Bootstrap of Pooled OOF Predictions

5,000 stratified bootstraps were performed on the pooled OOF predictions.

The result is:

Metric difference	Mean	Median	95% Bootstrap interval
AUROC	0.0459	0.0458	[0.0307, 0.0611]
Average Precision	0.0724	0.0724	[0.0491, 0.0948]

In 5,000 bootstrap resamples, the AUROC and AP differences for LightGBM were both greater than zero.

However, this bootstrap treats samples approximately as independent, does not repeat complete model training and nested hyperparameter selection, and does not fully reflect dependencies between cross-validated models. Therefore, it provides only supplementary descriptive evidence and cannot substitute for repeated nested cross-validation, patient-level bootstrapping, or independent external validation.

11. Interpretation of Results

When the following conditions are controlled:

- The same training data are used
- The same 12-hour sum features are used
- The same fixed outer-fold split is used
- The same primary evaluation metrics are used
- Parameters are selected only through inner-loop cross-validation
- Outer-fold labels do not participate in hyperparameter selection

LightGBM achieved higher AUROC and Average Precision in all five outer folds.

This result is consistent with the interpretation that under the current data and experimental settings, LightGBM's nonlinear modeling and feature interaction capabilities may be helpful in leveraging predictive information in high-dimensional sparse clinical features.

This is a reasonable explanation based on the model structure and experimental results, rather than a causal demonstration of the mechanism of the performance difference.

This experiment can only illustrate the relative performance of the model under the current retrospective data and verification framework, but cannot prove:

- LightGBM has reached clinical deployment requirements
- Model is able to maintain the same performance in other hospitals
- Model output probabilities have been fully externally calibrated in independent hospitals or independent time cohorts
- There is a causal relationship between important features and ARF
- The solution generated by RD-Agent must be better than all manual solutions

11.1 Supplementary Analysis of Probability Calibration and Threshold Policy

After the nested cross-validation was completed, this project further conducted cross-fitting probability calibration and threshold policy analysis on OOF predictions.

For each target outer fold:

- Fit the calibrator using the OOF predictions and labels of the remaining four outer folds;
- Use the remaining four folds to select the decision threshold;
- Apply calibrators and thresholds to target folds not involved in selection;
- Summarize the results of the five target folds.

Therefore, the target fold label does not participate in the calibrator fitting or threshold selection for that target fold.

The main calibration results are as follows:

Model	method	AUROC	AP	Brier score	Log loss	ECE	average predicted probability
Logistic Regression	Raw	0.7077	0.2288	0.1515	0.4824	0.1783	0.2687
Logistic Regression	Platt	0.7073	0.2277	0.0821	0.2928	0.0069	0.0968
LightGBM	Raw	0.7537	0.3017	0.0887	0.3030	0.0654	0.1585
LightGBM	Platt	0.7433	0.2917	0.0790	0.2808	0.0116	0.0988

The positive-class prevalence is 9.68%. Both models' raw predicted probabilities are systematically elevated. After Platt scaling, the average predicted probability is closer to the observed prevalence, and the Brier score, log loss, and ECE improve substantially.

Cross-fitting uses a different calibrator for each outer fold, so probability scales can change across folds. This may explain the modest decline in LightGBM's pooled OOF AUROC after Platt calibration. The situation differs from applying one monotonic transformation to all predictions from a single model.

Using the raw probabilities from LightGBM as an example, different threshold policies produce the following trade-offs:

threshold policy	Sensitivity	Specificity	Precision	Predicted positivity rate
Fixed 0.5	0.2594	0.9575	0.3952	0.0635
Youden J	0.6823	0.6843	0.1880	0.3512
Maximum F1	0.4292	0.8658	0.2551	0.1628
Target sensitivity 0.80	0.7844	0.5368	0.1535	0.4943
Target sensitivity 0.90	0.8917	0.3682	0.1313	0.6570

Improving sensitivity comes with a substantially higher burden of false positives and follow-up assessment. A fixed threshold of 0.5 is not inherently appropriate for this imbalanced task, and no single optimal threshold is independent of error costs and resource constraints.

This analysis remains an assessment of cross-fit within the same dataset and is not a substitute for independent hospital external calibration or prospective clinical validation.

The publicly released code, aggregate results, and limitations are described in Section 11.1 and the repository README. Individual-level OOF predictions and complete bootstrap draws are intentionally excluded from the public release.

12. Relationship with ARF V1

ARF V1 completed the following tasks:

- RD-Agent generation of prediction pipelines
- Human-developed logistic-regression baseline

- Fair model comparison
- Feature ablation
- Robustness analysis of multiple random seeds
- Feature importance analysis
- Threshold and error analysis

ARF V2 further strengthens the evaluation method on this basis:

- Fixing and preserving outer data partitions
- Recording SHA-256 hashes for data and results
- Nested cross-validation with separation between inner and outer loops
- Automated auditing of parameter-selection processes
- Verification through recalculation of OOF metrics
- Paired-difference and uncertainty analyses
- Cross-fitted probability-calibration and threshold-policy evaluation
- Clear distinction between mean fold-level metrics and pooled OOF metrics
- Clear distinction between the RD-Agent generation phase and the manual rigorous-review phase

Therefore, the focus of ARF V2 is not to simply obtain higher scores, but to improve the credibility, auditability, and reproducibility of experimental conclusions.

13. Limitations

13.1 Lack of External Validation

This experiment did not validate the model using data from independent hospitals or different time periods; therefore, the model's generalizability to other healthcare institutions or populations cannot be determined.

13.2 Missing Patient-Level Identifier

The data only provide ICU stay-level IDs, no patient-level identifiers. While each available ID appears only once, there is no way to further confirm whether records of multiple hospitalizations for the same patient were assigned across folds.

13.3 Limited Number of Outer Folds

This experiment has only five outer folds and therefore only five paired performance differences. The exact sign test has limited statistical power, so the five-fold result alone cannot support strong claims of statistical significance.

13.4 Lack of Repeated Nested Cross-Validation

This experiment uses a fixed five-fold outer-fold partition. Nested-cross-validation results under alternative random partitions have not been systematically replicated.

13.5 Limited Parameter Search Range

Logistic regression and LightGBM only select within a preset finite grid of parameters. The experimental results cannot represent the theoretical optimal performance of the two models under all possible hyperparameter configurations.

13.6 Feature Plan Based on Previous Results

The 12-hour summation features were chosen based on ablation and robustness experiments of ARF V1. Although ARF V2's model comparison uses strict nested parameter tuning, the feature scheme itself is not predetermined on a completely independent dataset.

13.7 Feature Anonymization

Feature names are represented as anonymous numbers, which limits interpretation of clinical significance and makes it difficult to map model importance to specific vital signs, test items, or treatment information.

13.8 Internal Cross-Fit Calibration Is Not a Substitute for External Calibration

This project has performed Platt scaling and isotonic regression using cross-fitting between outer folds and evaluated Brier score, log loss, ECE, calibrated intercept and calibrated slope.

However, these calibrators still come from OOF predictions within the same dataset. The model has not been externally calibrated in independent hospitals, independent time periods, or independent patient cohorts. Therefore, even if internal calibration metrics are improved, predicted probabilities cannot be directly interpreted as clinical occurrence probabilities that have been externally validated.

13.9 Bootstrap Limitations

The OOF Bootstrap in this experiment does not retrain the model and only reflects the changes in fixed OOF predictions under sample resampling, which cannot fully represent the uncertainty of the entire model development process.

13.10 Retrospective Study Limitations

This experiment is a retrospective machine learning study, not a prospective clinical trial, nor a medical device validation.

14. Final Conclusion

This ARF V2 experiment successfully established a rigorous evaluation process consisting of fixed partitioning, nested parameter selection, automatic result auditing, and statistical uncertainty analysis.

Under the same data, the same 12-hour sum features, and the same outer validation folds, LightGBM outperformed logistic regression in both AUROC and Average Precision across all five outer folds.

The main results are:

- Logistic regression mean outer-fold AUROC: 0.7080
- LightGBM mean outer-fold AUROC: 0.7758
- Mean paired AUROC improvement: 0.0677
- Corrected 95% interval: [0.0298, 0.1056]
- Logistic regression mean outer-fold AP: 0.2349
- LightGBM mean outer-fold AP: 0.3201
- Mean paired AP improvement: 0.0852
- Corrected 95% interval: [0.0154, 0.1549]

Both the five-fold results and the supplementary Bootstrap analysis consistently support LightGBM's better prediction performance under the current experimental conditions.

However, the exact sign test yielded `p = 0.0625` due to only five pairs of observations, and the study lacked patient-level identifiers, repeated nested cross-validation, and independent external validation. Therefore, this experiment does not claim to have demonstrated clinical validity or generalizability.

The rigorous conclusion that this experiment can support is:

> Under the current ARF data, with 12-hour sum features and a nested cross-validation framework, LightGBM demonstrates a performance advantage over logistic regression that is consistent in direction, practically meaningful in magnitude, and supported by multiple analyses. However, this conclusion still requires further validation through repeated nested cross-validation and independent external data.

The results of this study are only used for education, software reliability evaluation and machine learning method research, and cannot be used for actual medical diagnosis or treatment decisions.

15. Reproduction Method

Run all tests:

```
conda activate rdagent

python -m pytest -q
```

Run nested cross-validation:

```
python -m src.experiments.run_nested_cv
```

Run statistical uncertainty analysis:

```
python -m src.evaluation.uncertainty_analysis
```

Before reproducing, you need to use the environment variable `ARF_DATA_DIR` to point to the local training data directory containing the following files:

```
ARF_12h.csv

X.npz
```

If the data is saved in the project's original default directory, the loader will use the local default path.

Since the original medical data is not included in the research repository, external reproducers need to obtain the same data themselves from legitimate sources and confirm that the data versions are consistent using saved SHA-256 hashes.

16. Reproducible Results

Configuration

`configs/arf_nested_cv.json`

Fixed partition

The sample-level split file is generated locally and intentionally excluded from the public release. Its generation script and SHA-256 record are included.

Core programs

`src/data/arf_loader.py`

`src/experiments/run_nested_cv.py`

`src/evaluation/uncertainty_analysis.py`

Automated testing

`tests/test_splits.py`

`tests/test_arf_loader.py`

`tests/test_experiment_config.py`

`tests/test_nested_cv_outputs.py`

Result tables

`results/tables/nested_cv_fold_results.csv`

`results/tables/nested_cv_summary.csv`

`results/tables/nested_cv_selected_parameters.csv`

`results/tables/paired_fold_differences.csv`

`results/tables/paired_inference_summary.csv`

`results/tables/oof_bootstrap_summary.csv`

Excluded individual-level and execution records

The public release does not include the sample-level split file, OOF prediction file, full bootstrap-resample distribution, or complete execution logs.

Complete records

`data/checksums/arf_train_sha256.txt`

`data/checksums/arf_outer_folds_seed42.sha256`

`data/checksums/arf_nested_cv_v1.sha256`

`data/checksums/arf_uncertainty_v1.sha256`

`data/checksums/arf_final_english_reports.sha256`

17. Release Notes

Early exploratory work in ARF V1, human-developed baselines, feature ablation, multi-seed analysis, and feature-importance and threshold analyses are archived in `archive/arf_v1/`. ARF V2 is the primary version for this project's formal reporting and interpretation of results.

Part II. Probability Calibration and Threshold-Policy Analysis

1. Research Purpose

Using the out-of-fold (OOF) predictions generated by ARF V2 nested cross-validation, this analysis addresses the following questions:

- Do probabilities produced by logistic regression and LightGBM represent true ARF risk?
- Can Platt scaling and isotonic regression improve probability calibration?
- How do classification thresholds affect sensitivity, specificity, precision, and false-positive burden?
- How can a calibration method and threshold be selected without using labels from the target outer validation fold?

This experiment examines probabilistic reliability and threshold policy without retraining the underlying prediction models. It uses OOF predictions generated by nested cross-validation.

2. Input Data

The analysis uses nested-cross-validation out-of-fold predictions generated locally from the authorized task data. The individual-level prediction file is intentionally excluded from the public release. It contains 9,921 samples and the main fields include:

- `sample_index`
- `ID`
- `label`
- `outer_fold`
- `logistic_probability`
- `lightgbm_probability`

The label distribution is:

- Negative samples: 8,961
- Positive samples: 960
- Positive rate: 9.68%

Every sample has a complete OOF prediction, with no missing predicted probabilities. Each prediction is generated by an outer-fold model that was not trained on that sample.

3. Analysis and Design to Prevent Information Leakage

This experiment uses cross-fitting for probability calibration and threshold selection.

For each target outer fold:

- Fit the calibrator using the OOF predictions and labels of the remaining four outer folds;
- Select decision thresholds using only the remaining four outer folds;
- Apply calibrators and thresholds to target folds not involved in selection;

- Repeat this process for each of the five target folds;
- Summarize the results of all target folds.

Therefore, the target fold label does not participate in the calibrator fitting or threshold selection for that target fold.

This design is more stringent than fitting a calibrator or selecting a threshold on all OOF labels, and it can reduce optimism bias in decision analysis.

This remains a cross-fit evaluation within the same dataset; it is not equivalent to external validation in independent hospitals or cohorts.

4. Calibration Method

This experiment compares three probability-processing methods:

- `raw`: no additional calibration;
- `platt`: calibrates prediction scores using a parametric logistic mapping;
- `isotonic`: calibrates prediction scores using a monotonic nonparametric mapping.

Evaluation metrics include:

- AUROC;
- Average Precision;
- Brier score;
- Log loss;
- Expected Calibration Error (ECE) computed using 10 quantile bins;
- calibration intercept;
- calibration slope;
- Average predicted probability.

These metrics measure different aspects:

- AUROC measures the overall ranking ability of positive samples relative to negative samples;
- Average Precision places greater emphasis on positive-class detection quality under class imbalance;
- Brier score measures the mean squared error between probabilistic predictions and true binary outcomes;
- log loss imposes a greater penalty on confident but incorrect probability predictions;
- ECE measures the average difference between grouped predicted probabilities and observed event rates;
- The calibration intercept reflects whether the predicted probability is overall high or low;
- The calibration slope reflects whether the predicted probabilities are overly extreme or overly conservative;
- The average predicted probability can be compared with the overall positive-class prevalence.

AUROC and Average Precision mainly measure the ability to distinguish and sort; Brier score, log loss, ECE, calibration intercept and calibration slope mainly measure probabilistic reliability.

5. Probability Calibration Results

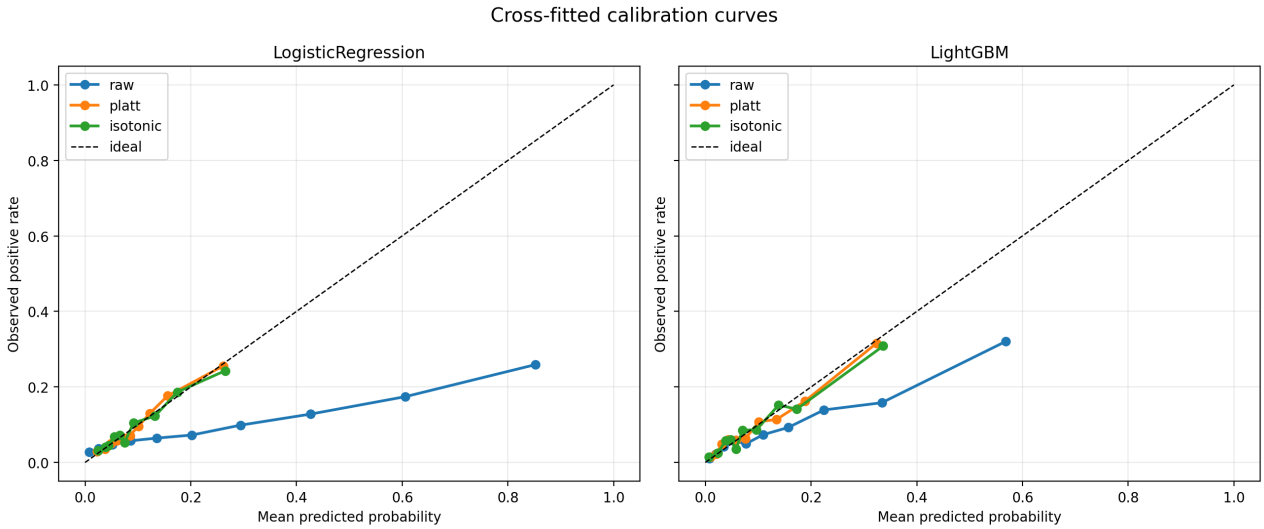


Figure 1. Cross-fitted probability calibration curves.

As part of the probability calibration results in Section 5, this figure compares the cross-fitted calibration results of logistic regression and LightGBM under raw probabilities, Platt scaling, and isotonic regression. For each target outer fold, the calibrator was fitted only using data from the remaining outer folds and then applied to the target fold that was not used for fitting. The figure compares agreement between predicted probabilities and observed event rates; it does not indicate external-hospital calibration, nor does it demonstrate that the model is ready for clinical deployment.

5.1 Logistic regression

method	AUROC	AP	Brier score	Log loss	ECE	average predicted probability
Raw	0.7077	0.2288	0.1515	0.4824	0.1783	0.2687
Platt	0.7073	0.2277	0.0821	0.2928	0.0069	0.0968
Isotonic	0.6988	0.2124	0.0826	0.2980	0.0107	0.0968

The raw average predicted probability from logistic regression is 26.87%, substantially higher than the observed positive rate of 9.68%. This suggests that the raw output of class-weighted logistic regression cannot be directly interpreted as the true event probability.

After Platt calibration:

- The average predicted probability decreased from 26.87% to approximately 9.68%;
- Brier score decreased from 0.1515 to 0.0821;
- log loss decreased from 0.4824 to 0.2928;
- ECE decreased from 0.1783 to 0.0069;
- AUROC remained largely unchanged.

These results show that Platt scaling substantially improves the probabilistic reliability of logistic regression in this cross-fit evaluation while largely preserving its ranking ability.

5.2 LightGBM

method	AUROC	AP	Brier score	Log loss	ECE	average predicted probability
Raw	0.7537	0.3017	0.0887	0.3030	0.0654	0.1585
Platt	0.7433	0.2917	0.0790	0.2808	0.0116	0.0988
Isotonic	0.7382	0.2789	0.0789	0.2840	0.0163	0.0990

The raw average predicted probability from LightGBM is 15.85%, which is also higher than the observed positive rate of 9.68%.

After Platt calibration:

- The average predicted probability decreased to about 9.88%;
- Brier score decreased from 0.0887 to 0.0790;
- log loss decreased from 0.3030 to 0.2808;
- ECE decreased from 0.0654 to 0.0116;
- Pooled OOF AUROC and AP both decreased modestly.

These results show that Platt scaling improves the probabilistic reliability of LightGBM, but its overall ranking metric decreases when all cross-fit predictions are combined.

5.3 Why Does the Pooled OOF AUROC Change After Calibration?

For a single model and a single dataset, strictly monotonic probability transformations generally do not change the AUROC.

However, this experiment uses cross-fit calibration:

- Use a different calibrator for each outer fold;
- Each calibrator is fitted by the remaining four folds;
- The mapping parameters of different folds may be different;
- There may be relative changes in the probability scale between different outer folds.

Therefore, even if the Platt mapping remains monotonic within each fold, the overall ranking of samples across the folds may still vary after pooling the calibrated probabilities across the five folds. This could explain the decrease in LightGBM pooled OOF AUROC from 0.7537 to 0.7433.

This is not the same as applying the same monotonic transformation to all predictions of a single model.

6. Calibration Method Selection

For LightGBM, isotonic regression achieved a slightly lower Brier score:

- Platt: 0.0790
- Isotonic: 0.0789

However, isotonic regression did not outperform Platt scaling on ECE, log loss, AUROC, or AP.

This choice also takes into account:

- Sample class imbalance;
- Only four outer folds are used per calibrator;
- The isotonic method has greater flexibility;
- isotonic regression is more susceptible to limited positive samples and local distributional fluctuations;
- Platt's ECE and log loss are better;
- Platt is simpler and easier to reproduce;

This experiment uses Platt scaling as the current preferred calibration method.

This choice is based on numerical performance, model simplicity, and stability in the current data. It does not imply that Platt scaling will remain optimal in independent hospital data.

This analysis did not calculate confidence intervals for metric differences between calibration methods; it therefore cannot support claims of statistically significant differences between methods.

7. Threshold Policy

The analysis compares five threshold policies:

- fixed threshold 0.5;
- Youden J;
- Max F1;
- Target sensitivity 0.80;
- Target sensitivity 0.90.

All adaptive thresholds are selected using only data outside the target fold.

A threshold policy answers the question, "At what probability level should a positive decision be triggered?" This differs from the overall ranking question answered by AUROC.

Threshold selection requires a trade-off between the following outcomes:

- False negatives;
- False positives;
- Sensitivity;
- Specificity;
- Precision;
- Predicted-positive rate;
- Resources for follow-up review or intervention.

8. LightGBM Threshold Results

The following results use the raw probabilities from LightGBM as an example:

threshold policy	average threshold	Sensitivity	Specificity	Precision	F1	Balanced accuracy	Predicted positivity rate
Fixed 0.5	0.5000	0.2594	0.9575	0.3952	0.3132	0.6084	0.0635
Youden J	0.1606	0.6823	0.6843	0.1880	0.2948	0.6833	0.3512
Maximum F1	0.3334	0.4292	0.8658	0.2551	0.3200	0.6475	0.1628
Target sensitivity 0.80	0.0952	0.7844	0.5368	0.1535	0.2568	0.6606	0.4943
Target sensitivity 0.90	0.0506	0.8917	0.3682	0.1313	0.2289	0.6299	0.6570

These threshold values apply only to the current raw probabilities from LightGBM.

After Platt or isotonic calibration, the probability scale changes, so the raw probability thresholds cannot be directly applied to the calibrated probabilities. Each probability representation requires reselection of the threshold using corresponding development data.

9. Threshold-Policy Interpretation

9.1 Fixed Threshold 0.5

A fixed threshold of 0.5 has high specificity, but identifies only about 25.9% of positive cases and therefore misses many cases.

After probability calibration, the sensitivity of the fixed 0.5 threshold may decrease further. This does not indicate calibration failure; rather, in data with a positive rate of about 9.68%, calibrated risk probabilities rarely exceed 50%.

Therefore, 0.5 is just a numerically common default threshold and not a clinical decision-making threshold that is inherently reasonable in this task.

9.2 Youden's J

Under the current evaluation, Youden's J yielded approximately:

- Sensitivity: 68.2%
- Specificity: 68.4%
- Precision: 18.8%
- Predicted positivity rate: 35.1%

It provides a more balanced sensitivity and specificity, but its predicted-positive rate is substantially higher than the observed positive rate, meaning that follow-up review is required for more patients.

Youden's J implicitly treats sensitivity and specificity as similarly important, whereas the costs of false negatives and false positives in real-world medical applications are often unequal.

9.3 Maximum F1

The maximum F1 policy results in:

- Sensitivity: 42.9%
- Specificity: 86.6%
- Precision: 25.5%
- F1: 32.0%

This policy improves precision and reduces the number of predicted positives, but has lower sensitivity. It places greater emphasis on the overall balance between precision and recall than on minimizing missed diagnoses to the greatest extent.

9.4 Target Sensitivity 0.80

To get close to 80% sensitivity on outer folds not involved in threshold selection:

- Actual sensitivity was approximately 78.4%;
- Approximately 49.4% of samples were classified as high risk;
- Specificity was approximately 53.7%;
- Precision was approximately 15.4%.

This means that nearly half of patients will trigger a high-risk flag, creating a substantial burden of follow-up review and intervention.

9.5 Target Sensitivity 0.90

To get close to 90% sensitivity:

- Actual sensitivity was approximately 89.2%;
- Approximately 65.7% of samples were classified as high risk;
- Specificity was approximately 36.8%;
- Precision was approximately 13.1%.

This policy reduces false negatives but generates a large number of false positives.

The actual outer sensitivity is slightly lower than the target value because the threshold is only selected on other folds. This reflects the ordinary error that arises when the threshold is generalized to unseen data, rather than a program error.

10. Distinctions Between Probability Calibration, Discrimination, and Decision Thresholds

Probability calibration, discrimination, and decision thresholds answer different questions:

- AUROC measures whether the model can rank positive samples ahead of negative samples;
- Average Precision measures the ranking quality of positive predictions under class imbalance conditions;
- Probability calibration measures whether a predicted risk level corresponds to a similar actual incidence;
- Threshold policies translate continuous risk probabilities into specific positive or negative decisions.

Therefore:

- A model can have a good AUROC but still have probabilities that are systematically high or low;

- As probabilistic calibration improves, there is no guarantee that AUROC will improve at the same time;
- The same model can produce completely different sensitivities and false positive burdens at different thresholds;
- There is no unique optimal threshold that is independent of specific usage scenarios and error costs.

11. Main Conclusions

- The raw probabilities of both logistic regression and LightGBM overestimate the risk of ARF;
- Platt scaling substantially improves Brier score, log loss, and ECE in the current cross-fit evaluation;
- The average predicted probability after Platt calibration is close to the observed positive rate;
- Under the current data and evaluation metrics, Platt scaling provides a more appropriate overall trade-off than isotonic regression;
- The pooled OOF ranking index decreased after LightGBM calibration, which may be related to the use of different calibration mappings for different outer folds;
- A fixed 0.5 threshold is not suitable for direct use as the default policy for this imbalanced risk-prediction task;
- Improving sensitivity requires bearing a substantially increased burden of false positives and follow-up review;
- A target-sensitivity policy cannot guarantee that the target set during development will be attained exactly on unseen outer folds;
- There is no unique optimal threshold value that is independent of application costs, resource constraints, and clinical safety requirements;
- Both calibrators and thresholds were selected via cross-fitting, without using target outer fold labels.

12. Limitations

- The lack of independent hospital data makes it impossible to evaluate external calibration;
- The lack of patient-level identifiers makes it impossible to completely rule out records from multiple hospitalizations of the same patient;
- Different outer folds use different calibrators, so the pooled OOF ranking may change;
- The isotonic method may be affected by the limited number of positive samples;
- ECE depends on the binning method and number of bins;
- The current analysis does not provide confidence intervals for differences in calibration metrics;
- Threshold policies do not incorporate true medical costs, bed capacity, manual review capacity, or intervention side effects;
- Target sensitivity policies were evaluated only in retrospective data;
- Current thresholds have not been prospectively validated;
- No decision curve analysis or net benefit analysis was performed in the current analysis;
- Calibration performance may vary across hospitals, time, populations, and disease incidence;
- This analysis does not demonstrate clinical effectiveness and cannot be used directly for diagnosis or treatment.

13. Reproducibility and Completeness

Calibration and threshold analysis scripts:

`src/evaluation/calibration_threshold_analysis.py`

Automated testing:

`tests/test_calibration_threshold_outputs.py`

Main output:

- `results/tables/calibration_metrics.csv`
- `results/tables/calibration_bins.csv`
- `results/tables/threshold_policy_fold_results.csv`
- `results/tables/threshold_policy_summary.csv`
- `results/figures/calibration_curves_v2.png`

Integrity check:

`data/checksums/arf_calibration_threshold_v1.sha256`

Public release boundary:

The released repository contains the calibration code, aggregate tables, figure, automated test, and checksum record. It does not contain individual-level OOF predictions or complete bootstrap-resample draws.

Automatic test results:

```
32 passed
```

14. Final Conclusion

Within a cross-fitting framework that does not use labels from the target outer validation fold, Platt scaling substantially improved the reliability of predicted probabilities for both logistic regression and LightGBM.

LightGBM retained stronger underlying discrimination, but whether predicted risk should be converted into an alert depends on the costs of missed diagnoses and false positives, review capacity, and practical resource constraints.

A fixed threshold of 0.5 produced low sensitivity in this task, whereas increasing the target sensitivity substantially increases the predicted-positive rate and false-positive burden. Thresholds must therefore be chosen according to a clear application objective and error costs, rather than determined solely by a single statistical metric.

The current results support the use of the model for research-oriented risk ranking, probability calibration, and threshold-policy comparison, but are insufficient to demonstrate clinical effectiveness or to support clinical deployment.

References

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Appendix: Figures from Early Exploratory Results

Figure A1. Feature Importance

The following figure comes from the ARF V1 exploratory analysis and shows the feature importances of the top 30 features ranked by the LightGBM model. Because the feature names are anonymized, the figure is mainly used to show model-usage patterns; it is not used for clinical or causal interpretation and is not a feature-selection result of the current formal nested cross-validation analysis.

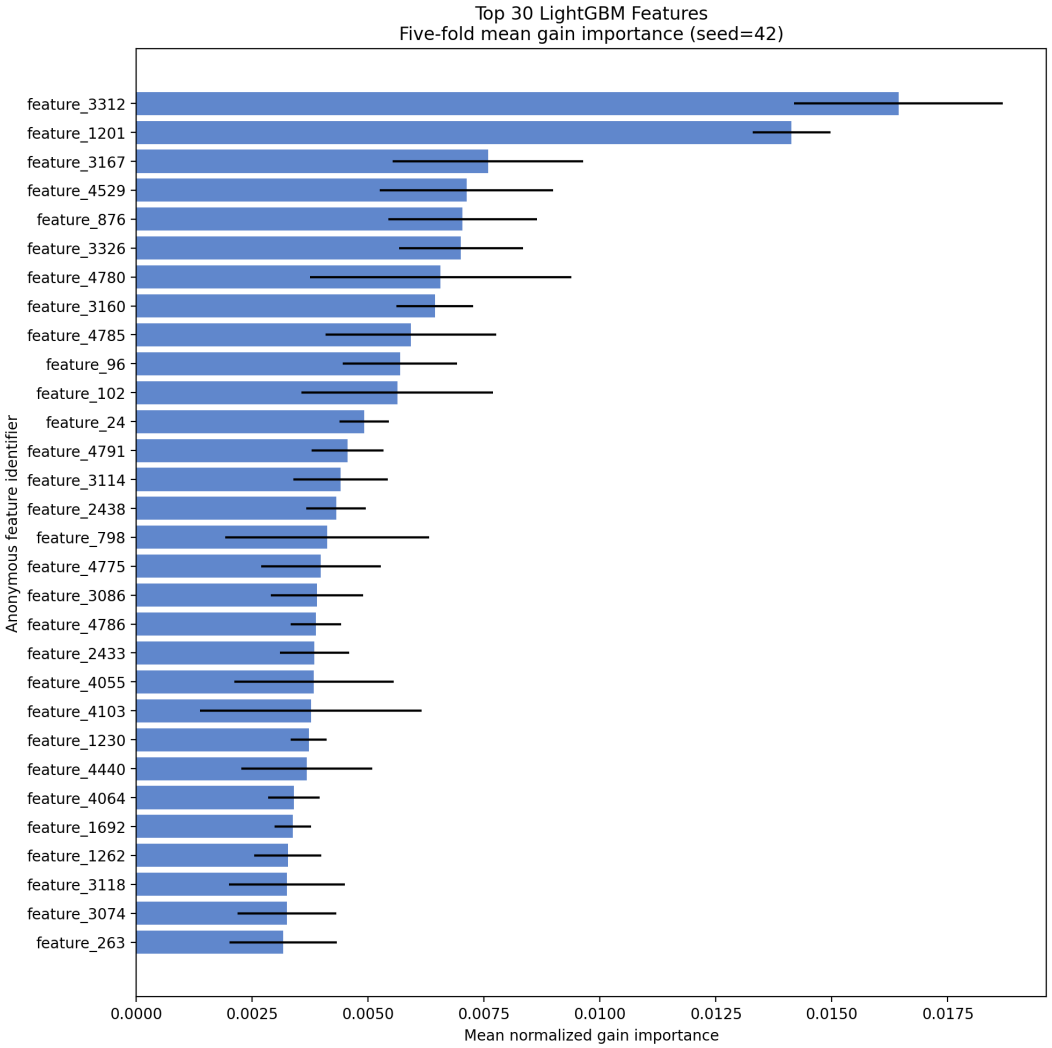


Figure A1. ARF V1 exploratory LightGBM top-30 feature importance.

Figure A2. Threshold and Error Analysis

The following figure is from the exploratory experiments in ARF V1 and presents sensitivity, specificity, precision, and error trade-offs at different decision thresholds. Formal conclusions about thresholds take as their basis the results obtained through the cross-fitting method in Part II.

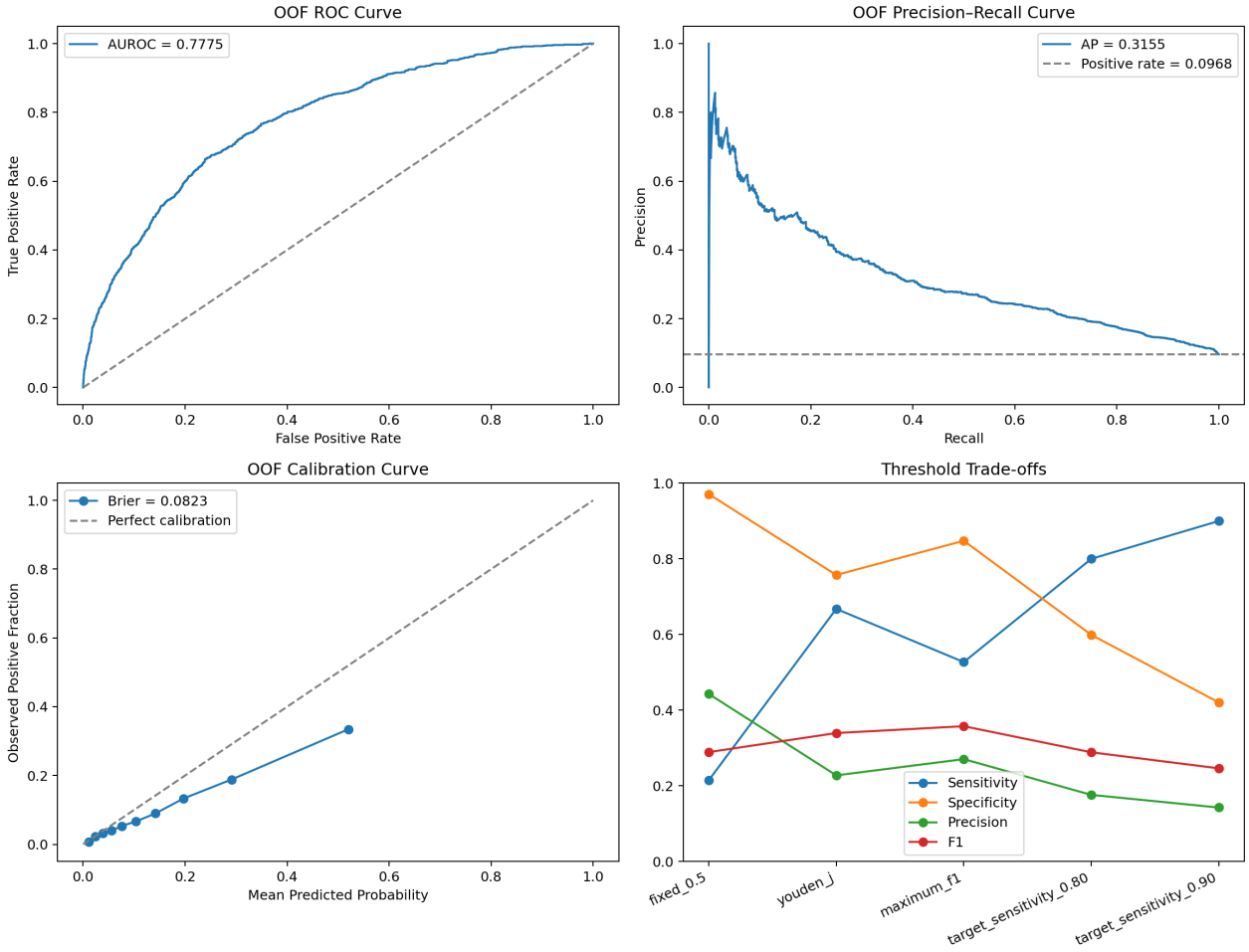


Figure A2. ARF V1 exploratory threshold and error analysis.